

- Putty. (PCs earlier than Windows 10 might need to use Putty. Do check references for installing Putty on MacOS.)
- Any third party SSH client, either open source or commercial.

Authentication

- Using `ec2-user@IP-Address` (or `ubuntu@IP-Address` for Ubuntu OS).
- Using the private key created in Step (2) of Web/middleware for Perl (using a password is not allowed in AWS).
 - `ssh -i "path\to\keypair.pem" useridForOS@IPAddress` (if using the command line).
 - If using Putty, then you have to convert the `.pem` file to a `.ppk` file using the Puttygen application by importing the `.pem` private key and saving it as a `.ppk` file. This can be imported into Putty's *Connection/SSH/Auth* section.

Upgrading Perl

Ubuntu comes with something that's very near the latest Perl. But all others come with unsupported Perl. So if we want to upgrade Perl, the string (as of this writing 5.28 is the latest).

This is done under the assumption that we have already run Step (3)(vi) of Web/middleware server for Perl and that we have a running Apache setup with the `mod_perl` module already installed.

It will do the following:

- Compile the downloaded `mod_perl` with the latest Perl.
- Overwrite the already existing `mod_perl` module.
- Reload Apache to activate the latest `mod_perl`.

```
#!/bin/bash
```

```
OSID=$(grep ^ID= /etc/os-release | awk -F= '{print $2}' | sed -e 's/"///g')
```

```
case $OSID in
```

```
amzn|rhel|fedora|centos)
```

```
    sudo yum -y groupinstall "Development Tools"
```

```
    sudo yum -y install httpd-devel
```

```
;;
```

```
sles)
```

```
    sudo zypper -n install --type pattern Basis-Devel
```

```
    sudo zypper -n install apache2-devel
```

```
;;
```

```
ubuntu|debian)
```

```
    sudo apt-get -y install build-essential
```

```
    sudo apt-get -y install apache2-dev
```

```
;;
```

```
esac
```

```
sudo mkdir -p /usr/local/src/{apache,perl}
```

```
wget -qO- http://apache.claz.org/perl/mod_perl-2.0.10.tar.gz | sudo tar xzv -C /usr/local/src/apache
```

```
wget -qO- https://www.cpan.org/src/5.0/perl-5.28.0.tar.gz | sudo tar xzv -C /usr/local/src/perl
```

```
cd /usr/local/src/perl/perl-5.28.0
```

```
sudo ./Configure -des -Duseithreads -Duseshrplib -Dprefix=/usr/local/perl5.28 && sudo make && sudo make install
```

```
cd /usr/local/src/apache/mod_perl-2.0.10
```

```
sudo /usr/local/perl5.28/bin/perl Makefile.PL MP_APXS=/usr/bin/apxs
sudo make && sudo make install #will install the module on top of the existing one
```

```
case $OSID in
```

```
amzn|rhel|fedora|centos)
```

```
    sudo systemctl restart httpd
```

```
;;
```

```
sles)
```

```
    sudo systemctl restart apache2 #SUSE
```

```
;;
```

```
ubuntu|debian)
```

```
    sudo service apache2 restart #Ubuntu
```

```
;;
```

```
esac
```

```
wget -q -O - http://localhost/cgi-bin/printenv |grep SERVER_SOFTWARE
```

The last line will print the Perl version. We can also open a browser and type the IP address from Step (4) of Web/middleware server for Perl. It should give a default page. Try the CGI path as well.

```
http://IPAddress/
```

```
http://IPAddress/cgi-bin/printenv
```

```
http://IPAddress/cgi-perl/printenv
```

You can see three or four more variables for the same script run from `mod_perl` compared to `mod_cgi`. We will discuss this in detail in the next article. Till then, happy hacking with Perl on the cloud! **END** 🐧

References

- <https://the.earth.li/~sgtatham/putty/latest/w64/putty-64bit-0.70-installer.msi>
- <https://www.youtube.com/watch?v=rblQ4y9coGg> – Putty on MAC
- <http://osxdaily.com/2017/04/28/howto-ssh-client-mac/>
- <https://www.i-programmer.info/programming/perl/9649-with-the-rise-of-devops-perl-shows-its-muscle.html>
- <https://alvinalexander.com/perl/edu/articles/pl020001.shtml>
- https://www.oreilly.com/library/view/practical-mod_perl/0596002270/ch02s04.html `mod_perl` on Windows
- <https://perl.apache.org/docs/1.0/guide/getwet.html> – Understanding `mod_perl`

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